

Binary Search – Recursive Method:

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#include <stdio.h>

int recursive_BinarySearch(int array[ ], int low, int high, int key)
{
    int mid;
    if (high >= low){
        mid = low + (high - low) / 2;
        if (array[mid] == key)
            return mid;
        if (array[mid] > key)
            return recursive_BinarySearch(array, low, mid - 1, key);
        return recursive_BinarySearch(array, mid + 1, high, key);
    }
    return -1;
}

int main( )
{
    int array[ ] = {1, 4, 7, 9, 16, 56, 70};
    int n = 7;
    int key = 9;
    int index = recursive_BinarySearch(array, 0, n - 1, key);
    if (index == -1) {
        printf("Element not found in the array ");
    }
    else {
        printf("Element found at index : %d", index);
    }
}
```

```
return 0;
```

```
}
```